CentraleSupelec

Software Engineering Project 2026

MySupermarket Checkout System

Project Report

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1. Introduction

This report covers the design and implementation of the MySupermarket checkout system, a Java application built for the Software Engineering Project 2026 at CentraleSupelec. The system handles a typical supermarket point-of-sale workflow: item scanning, bill computation with discount plans and category-level pricing, bank card payment through a POS/TAS pipeline, home delivery with 2-hour time-slot booking and dynamic pricing, and live stock tracking with low-stock alerts.

The application runs through a command-line interface (CLUI) as required by Part 2 of the project brief. All requirements R1 to R10 are implemented. The project is run from Eclipse (right-click Main.java, Run As, Java Application), or alternatively via the JAR at out/mysupermarket.jar.

2. Design Decisions

This section explains how we approached the main requirements and what design choices we made.

2.1 Facade: SupermarketSystem

We put all subsystem coordination inside one class called `SupermarketSystem`. This acts as a Facade so the CLI only needs to call methods on this one class and does not need to know how inventory, payment, or delivery work internally. It also made it easy to test individual subsystems in isolation.

2.2 Strategy pattern for discount plans (R5, R5b)

Discount plans are defined by a `DiscountPlan` interface with three methods: `discountMultiplier(subtotal)`, `annualFee()`, and `deliveryFeeMultiplier()`. There are three implementations: `NormalPlan`, `PrimePlan`, and `PlatinumPlan`. A new plan can be added by writing a new class implementing the interface and registering it in `PlanRegistry`. No existing code changes. This satisfies R5b.

2.3 Category pricing at runtime (R6, R6b)

Category discounts are stored in a `Map<String, Double>` inside `CategoryPricingPolicy`. When `setCategoryDiscount` is called with a new category name it just adds a map entry. There is no switch-case on category names, so R6b is satisfied without modifying existing classes.

2.4 Observer pattern for stock alerts (R9)

After each successful payment, `InventoryManager` decrements stock. If any item falls below its threshold, it calls `onLowStock(Item)` on every registered observer. At startup two `ManagerAlertObserver` instances are registered (one for Manager, one for Supplier). Adding

a new alert target just means implementing `StockObserver` and calling `addObserver()`. No other code changes.

2.5 Role checks in the CLI

The `CLI` class calls `requireRole(class, cmd)` before running each command, using Java's `instanceof` on the currently logged-in user. This keeps all access control logic in one place.

2.6 Delivery service (R7, R8, R8b)

We split delivery into two methods in `DeliveryService`: `computeRawFee()` applies the weight and distance rules, and `computeFee()` multiplies the result by `plan.deliveryFeeMultiplier()`. Any new plan with a different delivery discount just needs to return a different multiplier. The delivery service never needs to change.

2.7 Dynamic delivery slots (R10)

The `DeliverySlot` class models a 2-hour delivery window with a type (NORMAL, PEAK, ECO_FRIENDLY), a maximum vehicle capacity in kg, and a dynamic pricing multiplier (PEAK +30%, ECO_FRIENDLY -20%). `DeliveryService` pre-loads seven daily slots (S1 to S7). The customer uses `showSlots <weightKg>` to see available slots and `selectSlot <slotId>` to book one. The `bookSlot()` method prevents overbooking by checking remaining capacity before committing. The final delivery fee is: raw fee x plan multiplier x slot multiplier.

2.8 Payment flow (R4, Sections 2.5 and 2.6 of the spec)

The `POSTerminal` handles the full sequence: card lookup, PIN check, opening a TAS connection, authorisation, and debit. A `simulatedOutcome` field lets tests force any result (SUCCESS, INSUFFICIENT_FUNDS, PIN_WRONG, AUTH_DENIED) without touching bank state, which is used by the `simulatePayment` command.

2.9 Checkout as a snapshot

A `CheckoutSession` captures the customer, their plan, the active pricing policy, and any pending delivery address and slot ID at the moment `startCheckout` is called. Changes made by the manager during a checkout do not affect the current bill.

3. Design Patterns Used

Pattern	Where applied	Why
Facade	SupermarketSystem	Keeps the CLI simple. It only calls one class instead of reaching into all subsystems.
Strategy	DiscountPlan + NormalPlan / PrimePlan / PlatinumPlan	Each plan is its own class. New plans can be added without changing existing code. (R5b)
Observer	InventoryManager + StockObserver + ManagerAlertObserver	Stock alerts go to any number of listeners without the inventory knowing who they are. (R9)
Registry	PlanRegistry	Central lookup for discount plans. New plans can be registered at runtime. (R5b)
Open/Closed via Map	CategoryPricingPolicy	New category names accepted at runtime with no code changes. (R6b)

4. UML Diagrams

4.1 Class Diagram

The diagram below shows the main classes and their relationships. <<interface>> labels mark interfaces. Arrows show inheritance. Aggregation is shown with 'o'.

```
<<interface>>  DiscountPlan
  + getName(): String
  + discountMultiplier(subtotal: double): double
  + annualFee(): double
  + deliveryFeeMultiplier(): double
         ^               ^                ^
         |               |                |
   NormalPlan       PrimePlan        PlatinumPlan
   no fee,          50 EUR/yr,       200 EUR/yr,
   no discount      20% if >=50 EUR  30% always,
                    50% delivery     free delivery

PlanRegistry
  - plans: Map<String, DiscountPlan>
  + register(plan)  + get(name)  + all()

<<abstract>>  User
  - firstName, surname, username, password
  + checkPassword()  + getRole()
       ^          ^          ^
       |          |          |
   Manager    Cashier    Customer
                         - customerId: int
                         - address: String
                         - plan: DiscountPlan
                         - bankCard: BankCard
                         - pendingDeliveryAddress: String
                         - pendingDeliverySlotId: String   (R10)

SupermarketSystem  (Facade)
  - inventory: InventoryManager
  - planRegistry: PlanRegistry
  - pricingPolicy: CategoryPricingPolicy
  - deliveryService: DeliveryService
  - tas: TransactionAuthorisationSystem
  - pos: POSTerminal
  - users: Map<String, User>
  - currentSession: CheckoutSession
  - totalRevenue: double

DeliveryService
  - slots: List<DeliverySlot>     (R10)
  + computeRawFee()  + computeFee()  + bookSlot()  + getAvailableSlots()

DeliverySlot  (R10)
  - slotId: String
  - startTime / endTime: LocalTime
  - type: SlotType  { NORMAL, PEAK, ECO_FRIENDLY }
  - maxCapacityKg / currentLoadKg: double
  + hasCapacity()  + book()  + getPricingMultiplier()
    PEAK x1.30,  ECO_FRIENDLY x0.80,  NORMAL x1.00

InventoryManager
  - catalogue: Map<String, Item>
  - observers: List<StockObserver>
  + addObserver()  + decrementStock()  + restock()
```

```
       o  (notifies)
<<interface>>  StockObserver
  + onLowStock(item: Item)
       ^
       |
   ManagerAlertObserver
   - role: String  ("Manager" or "Supplier")
   + onLowStock(item)  ->  prints LOW STOCK alert

POSTerminal                          TransactionAuthorisationSystem
  - tas: TAS                           - cards: Map<String, BankCard>
  - simulatedOutcome                   - connectionOpen: boolean
  + pay(card, pin, amount)             + authorise(cardNum, amount): AuthResult
    : PaymentResult

CheckoutSession
  - customer: Customer
  - cart: List<CartEntry>
  - deliverySlotId: String  (R10)
  - pricingPolicy / deliveryService
  + addItem()  + computeBill()  + close()
```

4.2 Use Case Diagram

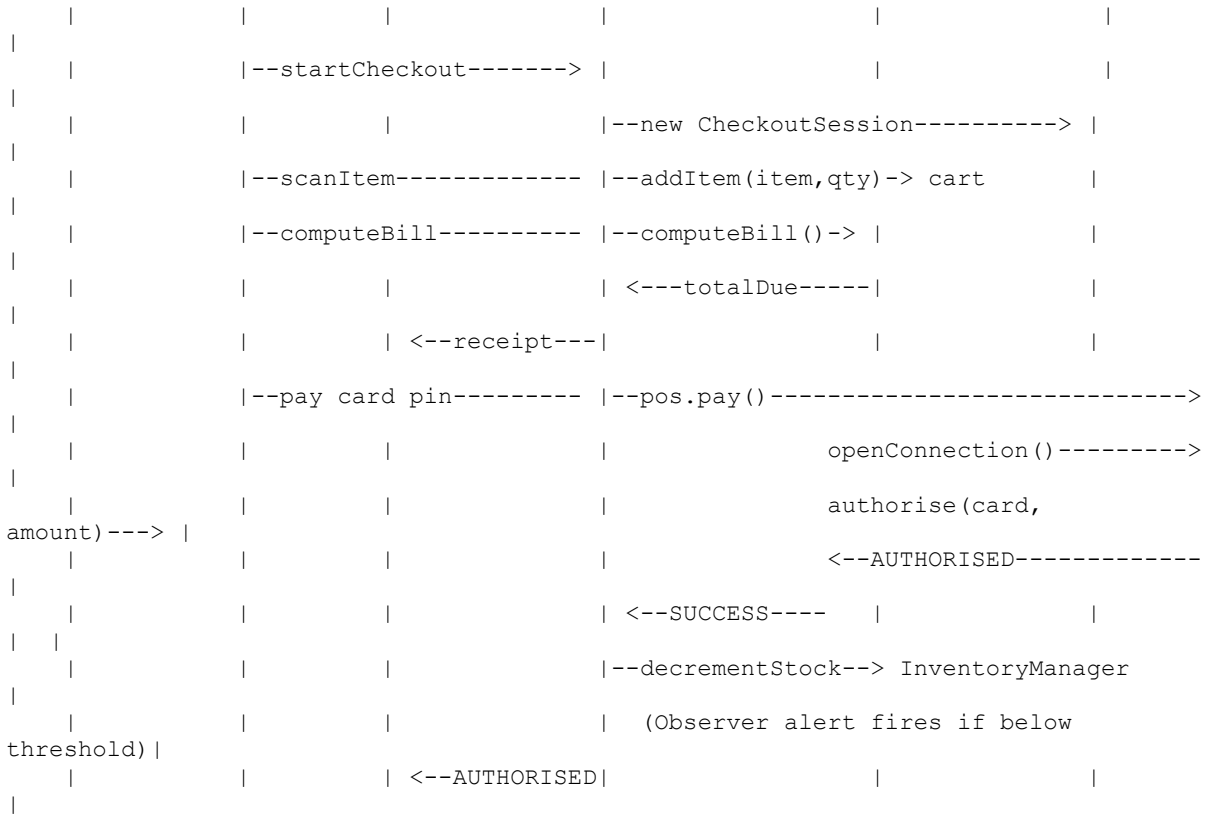
This shows which user types can run which commands.

```
+-------------------------------------------------------------------+
|  <<system>>  MySupermarket Checkout System                        |
|                                                                   |
|  Manager  -->  setup, registerCashier, registerCustomer           |
|           -->  addItem, restock, setCategoryDiscount              |
|           -->  showInventory, showRevenue                         |
|                                                                   |
|  Cashier  -->  startCheckout, scanItem, computeBill               |
|           -->  pay, simulatePayment                               |
|                                                                   |
|  Customer -->  subscribeToPlan, requestDelivery                   |
|           -->  showSlots, selectSlot          (R10)               |
|                                                                   |
|  TAS/Bank -->  authorise transaction  (called by POSTerminal)     |
|                                                                   |
|  All      -->  login, logout, help, runTest                       |
+-------------------------------------------------------------------+
```

4.3 Sequence Diagram: Successful Checkout and Payment

The sequence below shows the happy-path flow including slot selection and payment.

```
 Customer   Cashier     CLI      SupermarketSystem  CheckoutSession  POSTerminal
TAS
    |           |         |              |                 |              |
|
    |--showSlots 5.95---> |              |                 |              |
|
    |           |         |--showSlots-> | lists S1..S7    |              |
|
    |--selectSlot S5----> |              |                 |              |
|
    |           |         |--selectSlot->| slot S5 saved   |              |
|
```


startCheckout
new CheckoutSession
scanItem
addItem(item,qty)-> cart
computeBill
computeBill()->
totalDue
receipt
pay card pin
pos.pay()
openConnection()
authorise(card, amount)
AUTHORISED
SUCCESS
decrementStock--> InventoryManager
(Observer alert fires if below threshold)
AUTHORISED

5. Requirements Coverage

Req.	Description	Status	Key classes
R1	System handles customer purchases	Done	SupermarketSystem, CheckoutSession
R2	A purchase is a collection of bought items	Done	CartEntry, CheckoutSession
R2b	Item categories: fruit-and-vegetables, dairy, meat (and custom)	Done	Item, CategoryPricingPolicy
R3	Items have a price; price can depend on quantity	Done	Item, CartEntry.rawSubtotal()
R4	Payment by debit or credit card	Done	POSTerminal, BankCard, TAS
R5	Prime: 20% off when subtotal >= 50 EUR. Platinum: 30% always.	Done	PrimePlan, PlatinumPlan
R5b	New discount plans can be added without changing existing code	Done	DiscountPlan interface, PlanRegistry
R6	Category-level pricing policies (e.g. -10% on fruit-and-vegetables)	Done	CategoryPricingPolicy
R6b	New categories and policies can be added at runtime	Done	CategoryPricingPolicy (Map-based)
R7	Customer can request home delivery	Done	Customer, CheckoutSession
R8	Delivery fee based on total weight and distance. Orders over 50 kg refused.	Done	DeliveryService
R8b	Delivery discounts per plan: prime pays 50%, platinum pays nothing	Done	DiscountPlan.deliveryFeeMultiplier()
R9	Live stock tracking with Observer alerts when perishables go below threshold	Done	InventoryManager, StockObserver, ManagerAlertObserver
R10	2-hour delivery time slots, overbooking prevention, dynamic pricing (PEAK +30%, ECO_FRIENDLY -20%)	Done	DeliverySlot, DeliveryService.bookSlot(), showSlots, selectSlot

6. Design Analysis

6.1 What works well

- Both discount plans and category pricing can be extended without changing existing code, which directly satisfies R5b and R6b.
- The simulatedOutcome field in POSTerminal lets any payment outcome be tested without touching bank state, making it simple to write reproducible tests for failure cases.
- The CLI only calls SupermarketSystem, so it stays clean and does not need to know how the Observer chain, TAS, or delivery slots work.
- Adding a new stock observer only requires implementing StockObserver and calling addObserver(). Nothing else changes.
- Delivery slot booking prevents overbooking by checking remaining vehicle capacity before committing, and the dynamic pricing multiplier (PEAK/ECO_FRIENDLY/NORMAL) is fully integrated into the bill computation.
- Pricing policies and slot selections are captured at checkout start, so changes made by the manager mid-session do not affect the current bill.

6.2 Current limitations

- Everything is stored in memory. Restarting the application loses all data. Adding a file or database layer would fix this.
- Only one checkout session can be active at a time because SupermarketSystem holds a single currentSession field. Supporting multiple tills would need a map from cashier to session.
- The distance estimate in DeliveryService uses a simple character-sum formula rather than a real geocoding API.
- Passwords are stored as plain text. A real system would hash them.

7. Test Scenarios

7.1 Configuration file: my_supermarket.ini

This file loads automatically at startup. It sets up:

- Logs in as the default CEO manager.
- Runs setup, which loads the default catalogue (apple, banana, tomato, milk, yogurt, cheese, steak, chicken), two pre-registered test cards (4242424242424242 and 1111222233334444), a default cashier and default customer.
- Sets category discounts: fruit-and-vegetables -10%, dairy -5%.
- Adds extra items: orange, carrot (fruit-and-vegetables), butter (dairy), lamb (meat).
- Registers cashiers bob (bobpwd) and carol (carolpwd).
- Registers customers alice and pierre, each with an auto-generated bank card with 1000 EUR balance.
- Logs out.

7.2 Test scenario 1 (testScenario1.txt) - end-to-end happy and sad path

This covers all mandatory requirements. Run it with:

```
runTest testScenario1.txt
```

Steps:

- Manager logs in, runs setup, then adds items including steak with initial stock 5 (threshold also 5).
- -10% category discount applied to fruit-and-vegetables.
- Alice logs in, subscribes to prime (50 EUR annual fee deducted immediately), and requests home delivery to '12 rue de la Paix'.
- Cashier bob opens checkout for alice. Items scanned: 2x milk, 3x yogurt, 1x steak, 10x apple, 4x tomato.
- computeBill: -5% on dairy (milk 2.28, yogurt 2.28), no discount on steak (12.50), -10% on fruit (apple 2.70, tomato 3.24). Subtotal = 23.00 EUR. Prime plan does not fire (below 50 EUR). Delivery: weight 5.95 kg, distance 12 km, flat fee 15.00 EUR x 0.5 (prime) = 7.50 EUR. Total due: 30.50 EUR.
- First payment: simulatePayment INSUFFICIENT_FUNDS, pay -> refused.
- Second payment: simulatePayment SUCCESS, pay -> authorised. Stock decremented: steak goes to 4, below threshold 5. Observer fires alerts to Manager and Supplier.
- Second checkout: 4x steak = 50.00 EUR, prime fires (20% off) -> total 40.00 EUR. Payment SUCCESS.
- Manager: showInventory shows steak with [LOW STOCK] marker, showRevenue shows 70.50 EUR.

7.3 Expected outputs from test scenario 1

What to check	Expected output
Delivery fee line in bill (checkout 1)	Delivery fee charged: 7.50 EUR
Bill total for checkout 1	TOTAL DUE: 30.50 EUR
Failed payment attempt	[POS] Payment REFUSED: insufficient funds.
Observer alert after checkout 1 (steak = 4)	[ALERT -> Manager] LOW STOCK: 'steak' - only 4 unit(s) remaining
Observer alert after checkout 2 (steak = 0)	[ALERT -> Manager] LOW STOCK: 'steak' - only 0 unit(s) remaining
Bill total for checkout 2 (prime fires)	TOTAL DUE: 40.00 EUR
steak in showInventory	steak ... 0 [LOW STOCK]
showRevenue at end	Total revenue: 70.50 EUR

7.4 Test scenario 2 (R10): dynamic slot pricing

This short scenario tests the slot booking and dynamic pricing from R10. It can be added to testScenario1.txt or run as a separate file.

```
# After the ini has loaded (alice registered, prime plan active)
login alice alicepwd
requestDelivery "12 rue de la Paix"
showSlots 5.95
# Expected output: lists S1-S7 with their type and remaining capacity
# S5 is 16:00-18:00, ECO_FRIENDLY, 200 kg remaining
selectSlot S5
# Expected: [INFO] Delivery slot S5 (16:00-18:00, ECO_FRIENDLY) selected for Alice
Martin.
logout

login bob bobpwd
startCheckout alice
scanItem milk 2
scanItem steak 1
computeBill
# Expected: raw fee = 15.00, prime multiplier 0.5 -> 7.50, ECO_FRIENDLY 0.80 ->
6.00
# Item subtotal 2*1.20*0.95 + 12.50 = 14.78 (below 50 EUR, prime does not fire)
# TOTAL DUE: 14.78 + 6.00 = 20.78 EUR
simulatePayment SUCCESS
pay 4242424242424242 1234
logout
```

Key thing to verify: *the slot S5 ECO_FRIENDLY multiplier (x0.80) reduces the delivery fee below the normal prime rate. Choosing slot S2 (PEAK, x1.30) would give 7.50 x 1.30 = 9.75 EUR instead.*

8. How to Run and Test the Implementation

8.1 Running the application

The primary way to run the project is from Eclipse:

- Open the project in Eclipse (File -> Import -> Existing Projects into Workspace).
- Right-click on src/cli/Main.java in the Package Explorer.
- Click Run As -> Java Application.
- The CLUI starts in the Console tab. The my_supermarket.ini file loads automatically if it is in the project root folder.

To run a test scenario from inside the application:

```
runTest testScenario1.txt
```

Alternatively, a compiled JAR is provided at out/mysupermarket.jar and can be run from the terminal:

```
java -jar out/mysupermarket.jar

# Non-interactive:
java -jar out/mysupermarket.jar < testScenario1.txt
```

8.2 Running JUnit tests

JUnit 5 tests are provided for all non-trivial methods. To run them in Eclipse: right-click the project -> Run As -> JUnit Test. All test classes are in the same package as the class they test. Test classes:

- ItemTest: stock decrement, restock, threshold flag.
- DiscountPlanTest: PrimePlan threshold (below and above 50 EUR), PlatinumPlan unconditional discount, annual fees.
- CategoryPricingPolicyTest: multiplier calculation, unknown category returns 1.0.
- DeliveryServiceTest: 50 kg refusal, flat fee, heavy and distance surcharges, plan multiplier, slot dynamic pricing.
- InventoryManagerTest: catalogue operations, Observer notification on low stock.
- BankTest: card not found, wrong PIN, insufficient funds, successful payment.
- SupermarketSystemTest: full checkout cycle, subscribe-to-plan fee charge, scanItem stock check, slot booking.
- CLITest: command routing, role enforcement, syntax error handling.

9. Workload Distribution

The table below shows who had primary responsibility for each component. Both students contributed to all phases.

Class / Task	Design	Code	UML Diagrams	JUnit Tests	Lines ~
SupermarketSystem (Facade)	Utkarsh	Utkarsh	Both	Shriya	200
DiscountPlan + Plans + PlanRegistry	Both	Shriya	-	Utkarsh	130
CategoryPricingPolicy	Shriya	Shriya	-	Utkarsh	50
InventoryManager + Observer classes	Utkarsh	Utkarsh	Both	Shriya	100
DeliveryService + DeliverySlot (R10)	Shriya	Shriya	Utkarsh	Both	150
POSTerminal + TAS	Utkarsh	Utkarsh	Both	Shriya	120
CLI + Main	Both	Both	-	Both	320
Model classes (Item, CartEntry, BankCard)	Shriya	Shriya	-	Utkarsh	130
Store classes (User, Customer, Cashier, Manager, CheckoutSession)	Utkarsh	Utkarsh	-	Shriya	220
Report	Both	-	Shriya	-	-
Test files (.ini, testScenario1.txt)	Both	Utkarsh	-	-	80

Total source lines excluding tests: approximately 1,600. Test source lines: approximately 650.

10. Conclusion

The MySupermarket system covers all requirements from R1 to R10. The design uses a Facade to keep the CLI simple, Strategy for discount plans, Observer for stock alerts, and a runtime-extensible Map for category pricing. R10 is fully implemented: DeliverySlot models 2-hour windows with PEAK, NORMAL, and ECO_FRIENDLY types, bookSlot() enforces vehicle capacity limits, and dynamic pricing is applied on top of the plan discount when the bill is computed.

The system was tested through the CLUI using testScenario1.txt and interactively. JUnit tests cover all the main logic. The project runs from Eclipse with a single right-click, and a compiled JAR is available for non-Eclipse users.